

CLASSIFICATION

CLASS IMBALANCE PROBLEM

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Associate Professor

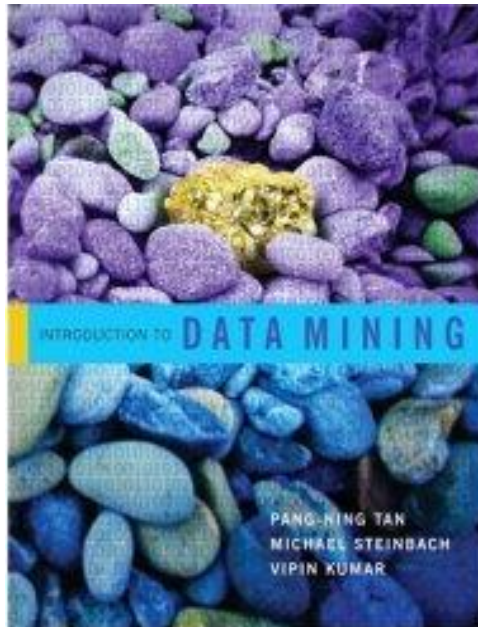
c/o Department of Informatics, Systems and Communication

University of Milano Bicocca



CLASS IMBALANCE PROBLEM

Part of the material presented in this lecture is taken from the following book.



Pang-Ning Tan, Michael Steinbach and Vipin Kumar
(2006). *Introduction to Data Mining*, Pearson
International.

Transcription and interpretation errors are responsibility of the lecturer.

CLASS IMBALANCE PROBLEM

The following concepts will be introduced:

- ✓ **CLASS IMBALANCE**
- ✓ **RARE CLASS**
- ✓ **POSITIVE/NEGATIVE CLASS**
- ✓ **PRECISION**
- ✓ **RECALL**
- ✓ **F₁-MEASURE**

CLASS IMBALANCE PROBLEM

1

Account Length	VMail Message	Day Mins	Churn	Intl Calls	Intl Charge	State	Area Code	Phone
128	25	265,1	n	3	2,7	KS	415	382-4657
107	26	161,6	n	3	3,7	OH	415	371-7191
137	0	243,4	n	5	3,29	NJ	415	358-1921
84	0	299,4	y	7	1,78	OH	408	375-9999
75	0	166,7	n	3	2,73	OK	415	330-6626
118	0	223,4	n	6	1,7	AL	510	391-8027
121	24	218,2	n	7	2,03	MA	510	355-9993
147	0	157	n	6	1,92	MO	415	329-9001
117	0	184,5	n	4	2,35	LA	408	335-4719
141	37	258,6	n	5	3,02	WV	415	330-8173

After the first Data Mining analysis of the **CUSTOMERS CALLS DATABASE** you notice that the number of **CHURNERS**, i.e. **UNFAITHFUL CUSTOMERS** who abandon the Telecommunication Company, are fortunately for your friend, the CEO, a **SMALL MINORITY**.



In particular, out of **3,333 CUSTOMERS** (a sample of the Telecommunication Company customers) **483 ARE CHURNERS** (14.5%).

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What about a **CLASSIFICATION MODEL ACHIEVING AN ACCURACY** value equal to **85.5%**?

A **CLASSIFICATION MODEL WHICH LABELS ALL THE 3,333 RECORDS** of the data set **AS NON CHURNER** achieves an **ACCURACY** value equal to **85.5%**.



Is such a **CLASSIFICATION MODEL USEFUL**?

It performs as the **ZERO RULE**.

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What about a **CLASSIFICATION MODEL ACHIEVING AN ACCURACY** value equal to **85.5%**?



How to proceed?



It performs as the **ZERO RULE**.

The **ACCURACY** measure treats every class as equally important, it may **not be suitable for analyzing IMBALANCED DATA SETS**, where the **RARE CLASS** is considered **more interesting than the MAJORITY CLASS**.

For **BINARY CLASSIFICATION**, the **RARE CLASS** is often denoted as **POSITIVE CLASS** while the **MAJORITY CLASS** is denoted as **NEGATIVE CLASS**.

We already introduced the concept of **CONFUSION MATRIX**.

		INDUCER PREDICTION (IP)	
		-1	+1
		TN	FP
ACTUAL CLASS (AC)	-1	TN	FP
	+1	FN	TP

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✓ **TN, true negative**, # of records where **AC=-1** which are correctly predicted **IP=-1**

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ACTUAL CLASS (AC)	-1	TN	FP
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✓ **FP, false positive**, # of records where **AC=-1** which are wrongly predicted **IP=+1**

The counts in a **CONFUSION MATRIX** can also be expressed in terms of percentages:

✓ **TNR, TRUE NEGATIVE RATE** or **SPECIFICITY**

Fraction of negative records predicted correctly by the Classification Model.

$$\text{TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

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✓ **FNR, FALSE NEGATIVE RATE**

Fraction of positive records predicted as a negative class by the Classification Model.

$$\text{FNR} = \frac{\text{FN}}{\text{TP} + \text{FN}}$$

RECALL and **PRECISION** are two widely used metrics employed in applications where detection of one of the classes is considered to be more important than detection of the other classes.

$$\text{Precision, } p = \frac{TP}{TP + FP}$$

PRECISION determines the **FRACTION OF RECORDS THAT ACTUALLY TURNS OUT TO BE POSITIVE IN THE GROUP THE CLASSIFICATION MODEL HAS DECLARED AS A POSITIVE CLASS.**

The **HIGHER** the **PRECISION** is, the **LOWER** the **NUMBER OF FALSE POSITIVE** errors committed by the Classification Model.

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$$\text{Recall, } r = \frac{TP}{TP + FN}$$

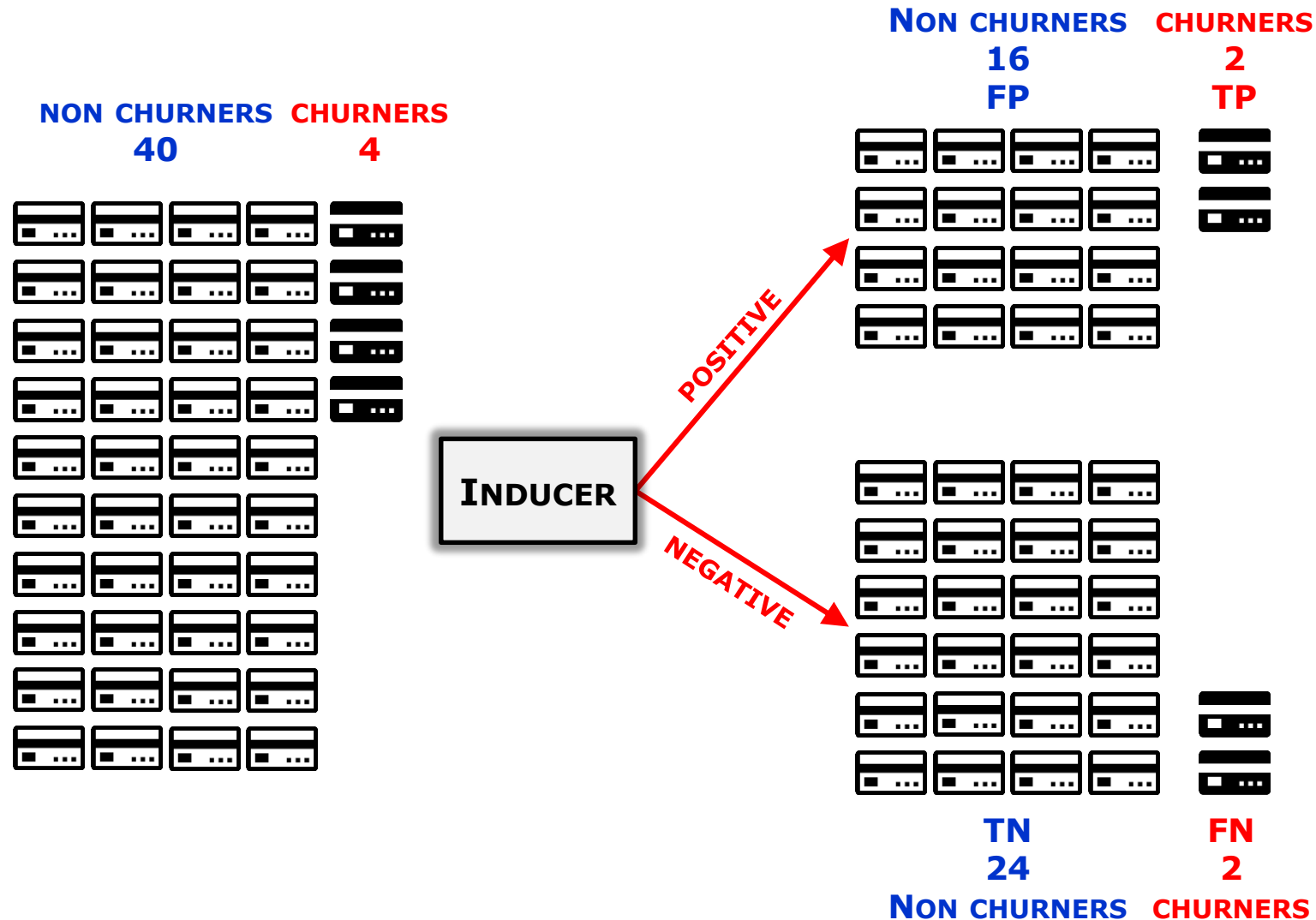
RECALL measures the **FRACTION OF POSITIVE RECORDS CORRECTLY PREDICTED BY THE CLASSIFICATION MODEL.**

LARGE RECALL means very **FEW POSITIVE RECORDS MISCLASSIFIED AS NEGATIVE CLASS.** In fact, **RECALL IS EQUIVALENT TO THE TRUE POSITIVE RATE (TPR).**

CLASS IMBALANCE PROBLEM: PRECISION-RECALL

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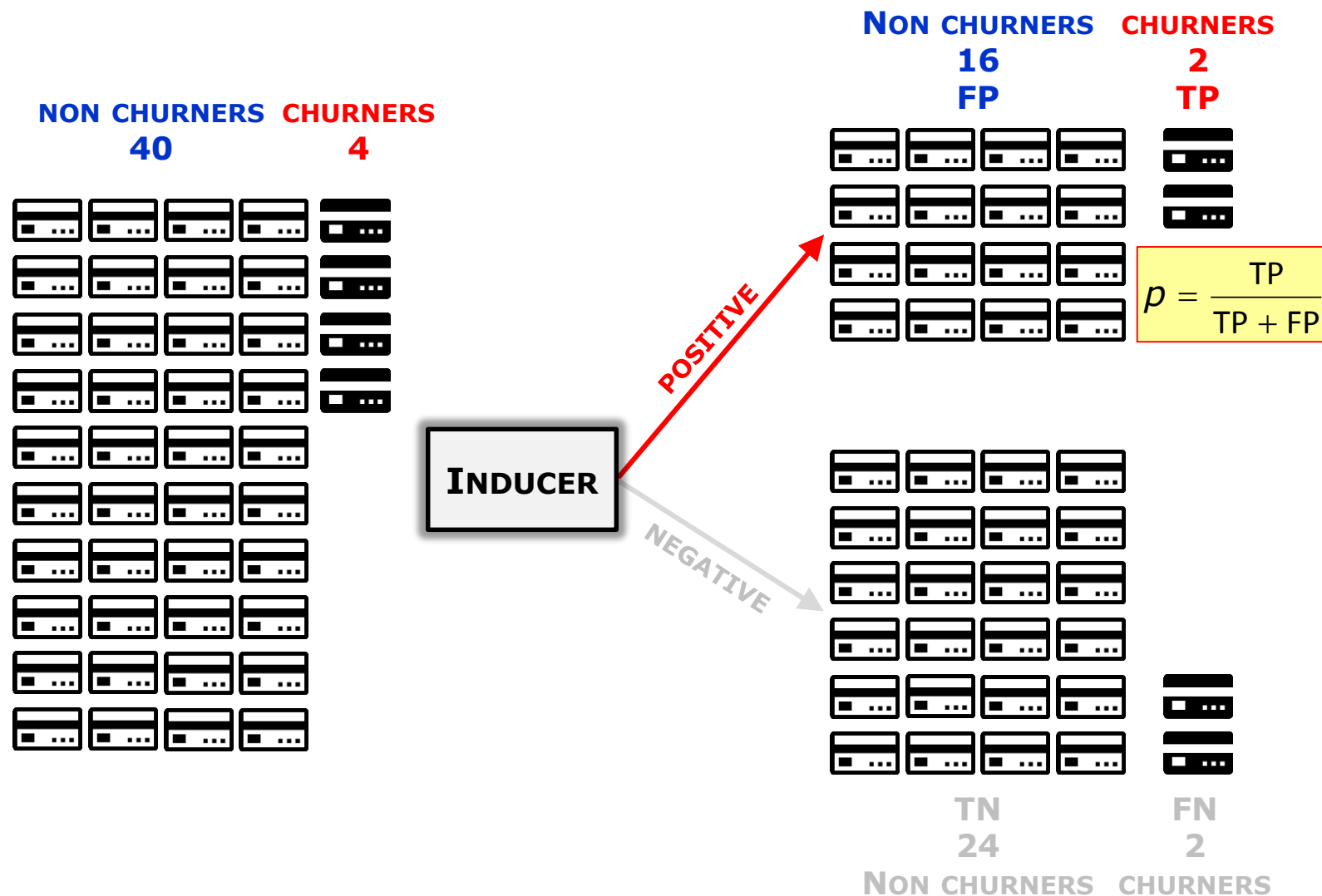
Back to the **CHURN PROBLEM**, assume the following:



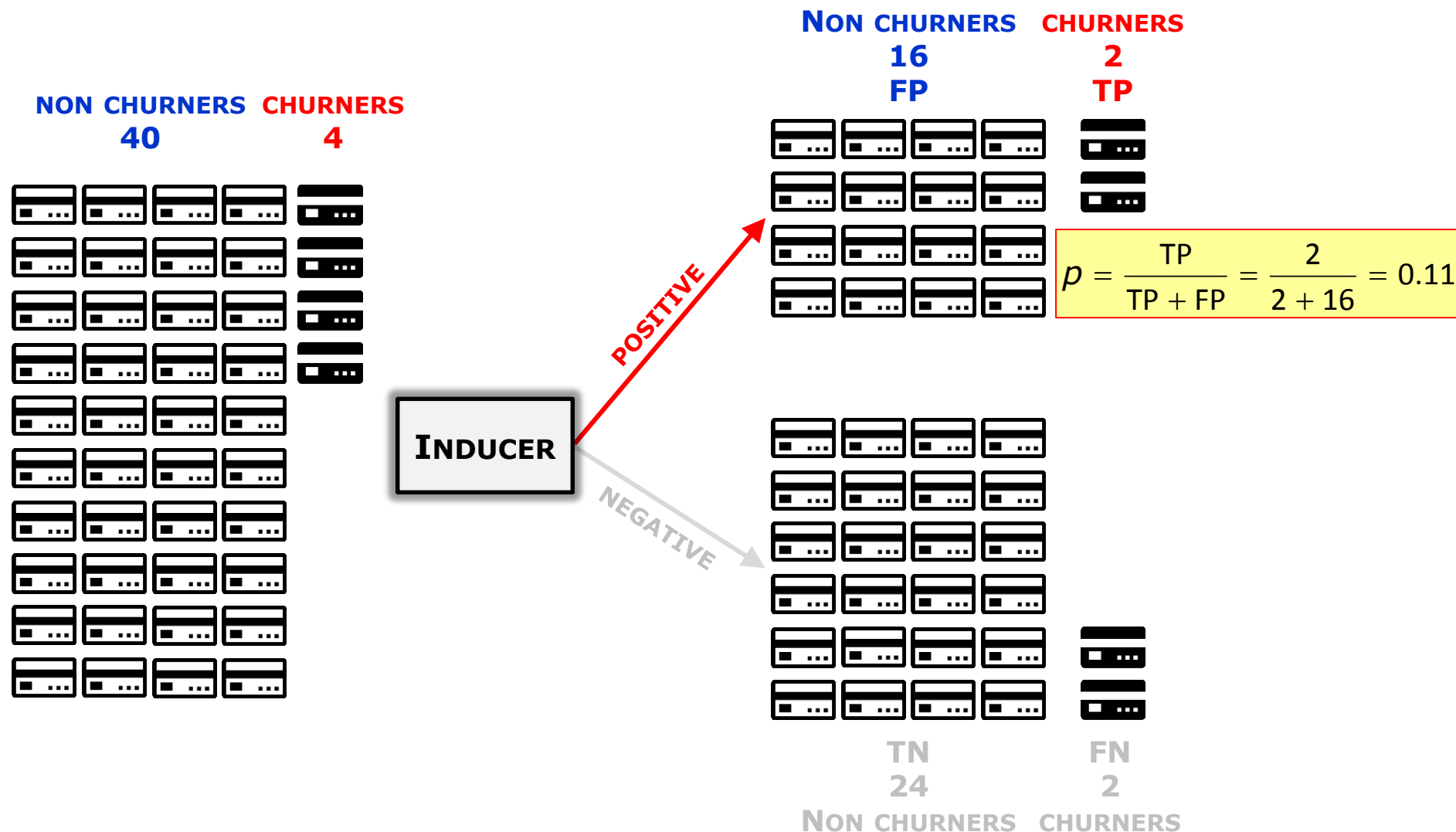
CLASS IMBALANCE PROBLEM: PRECISION-RECALL

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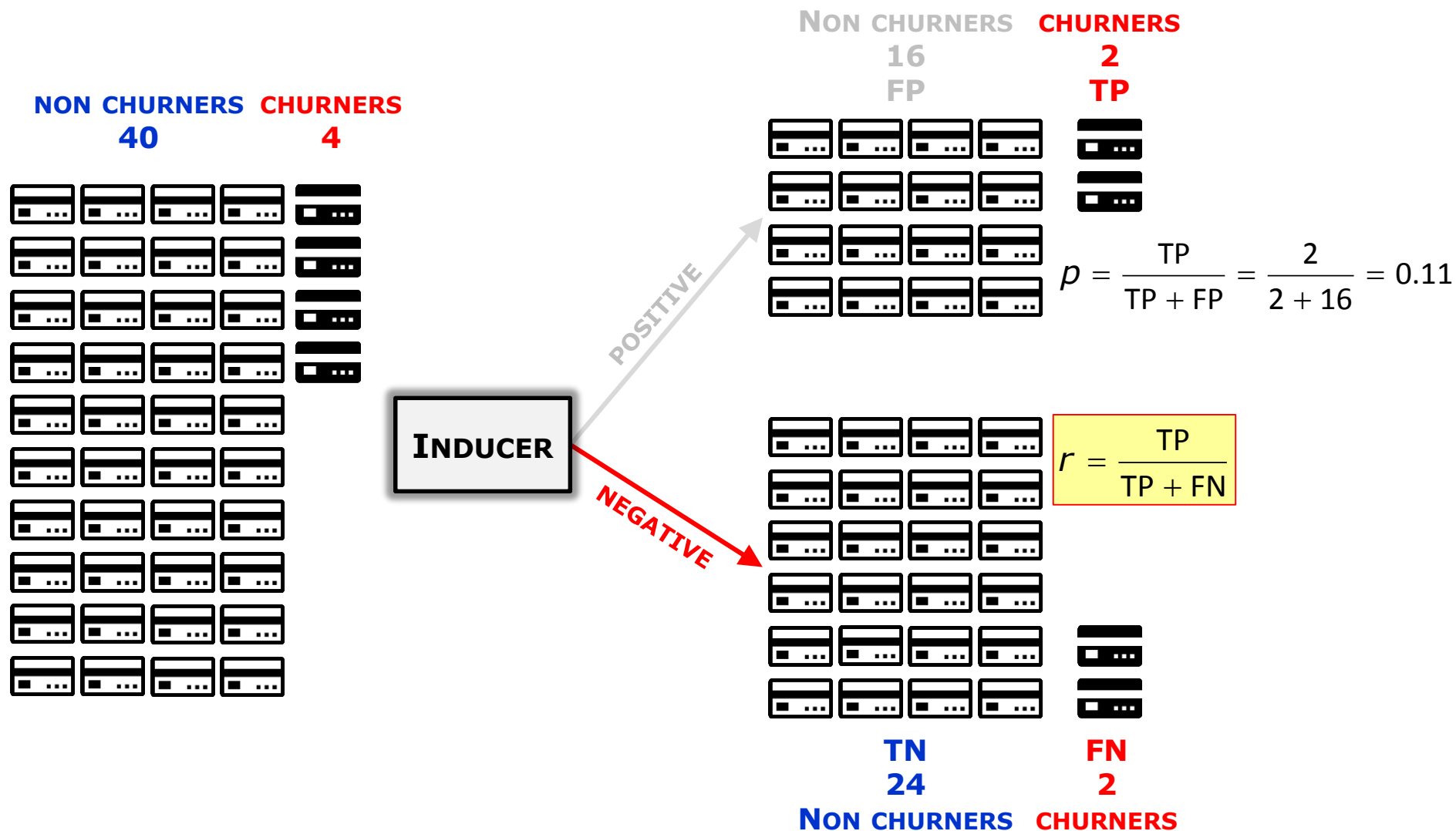
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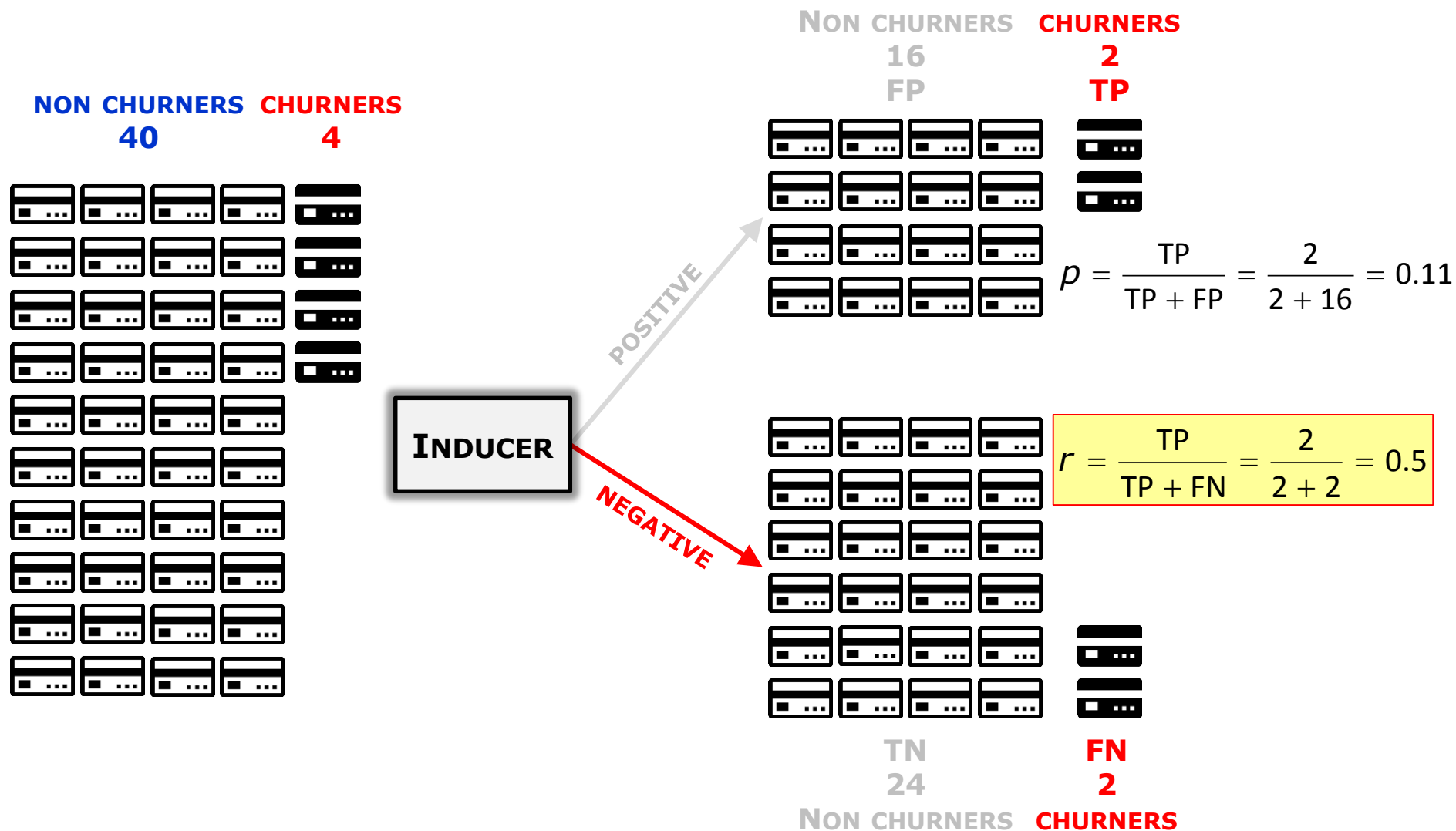
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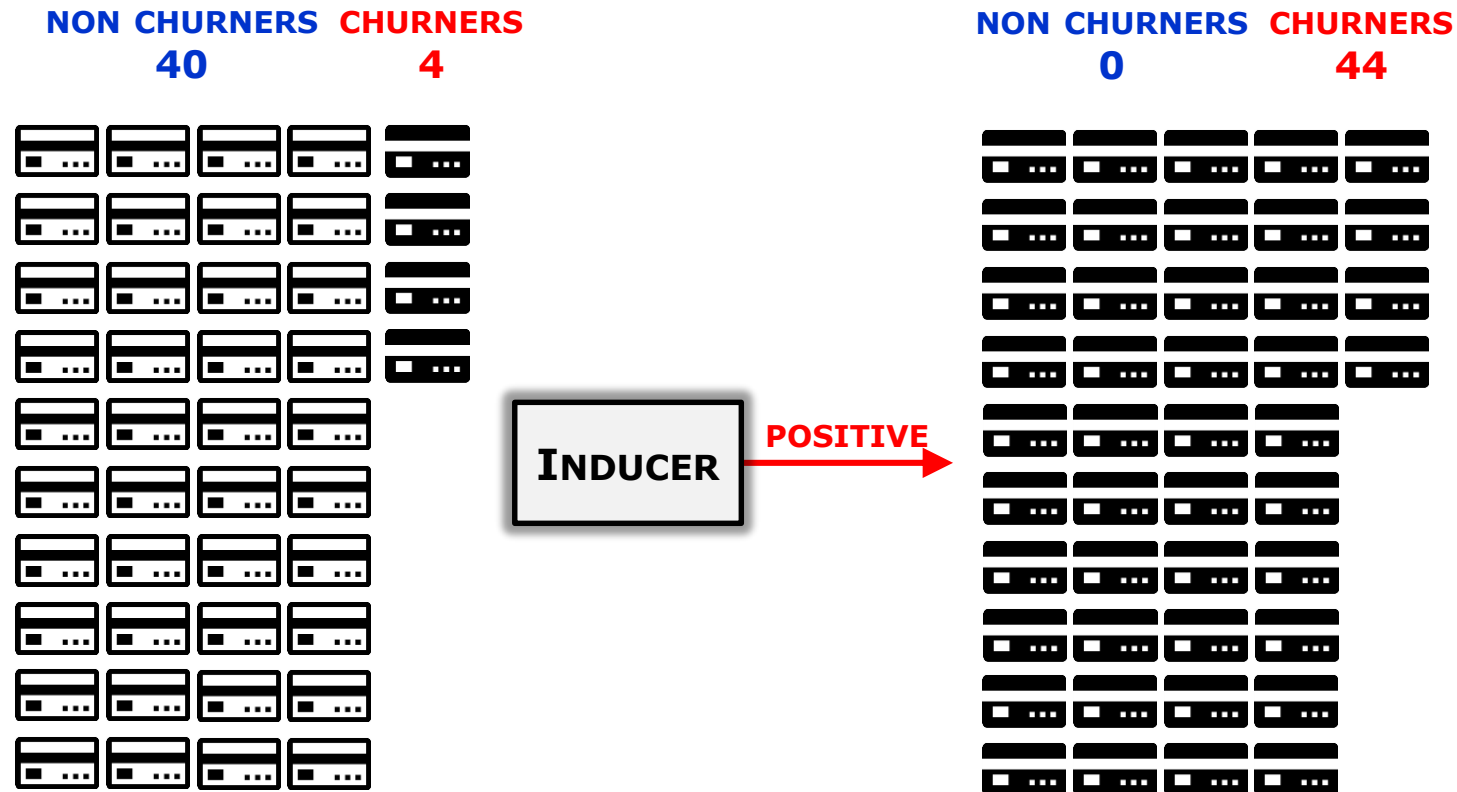
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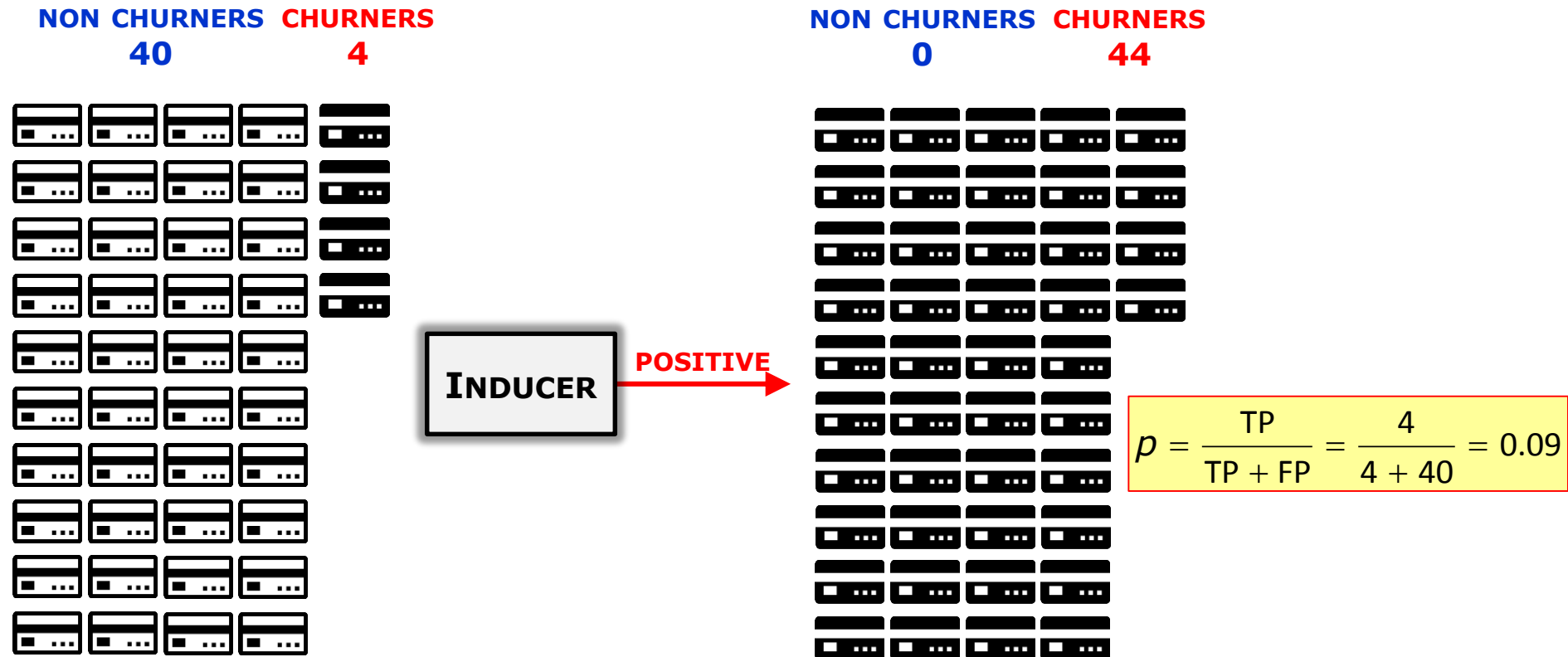
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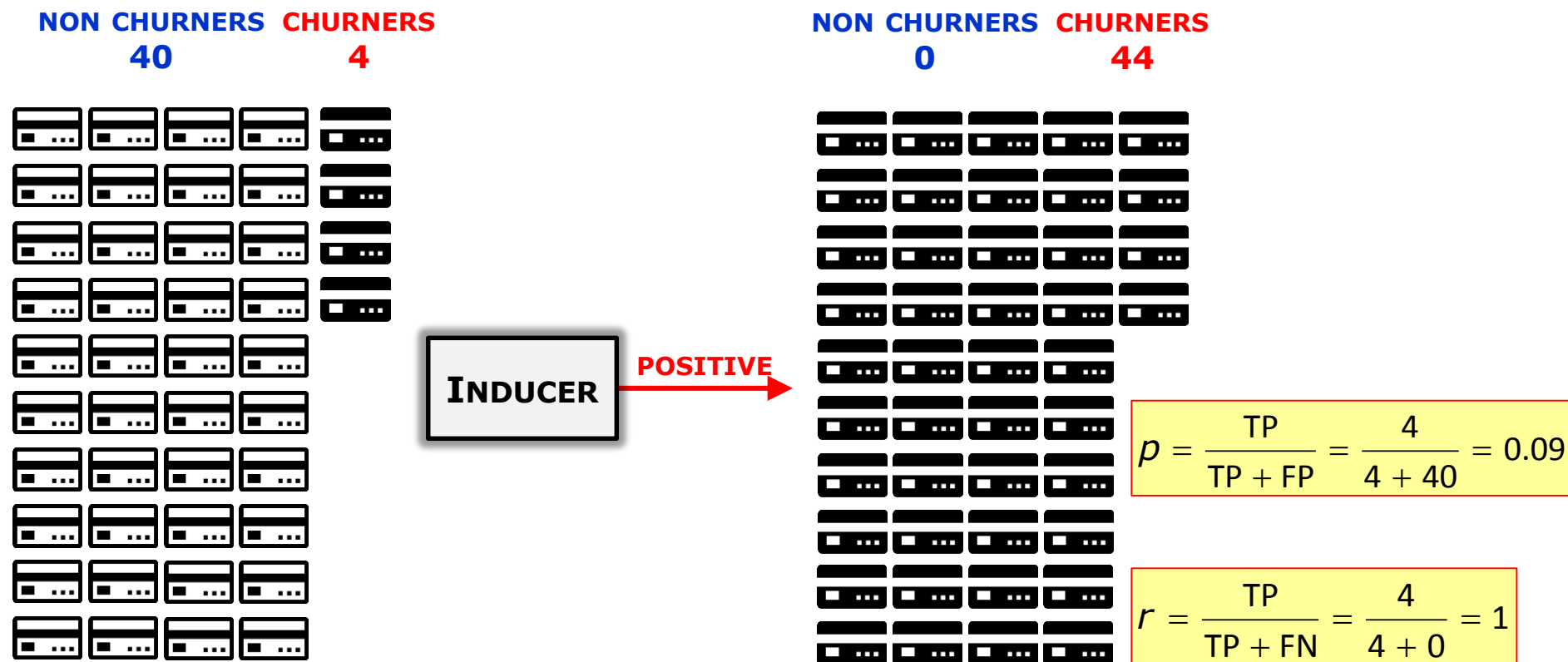
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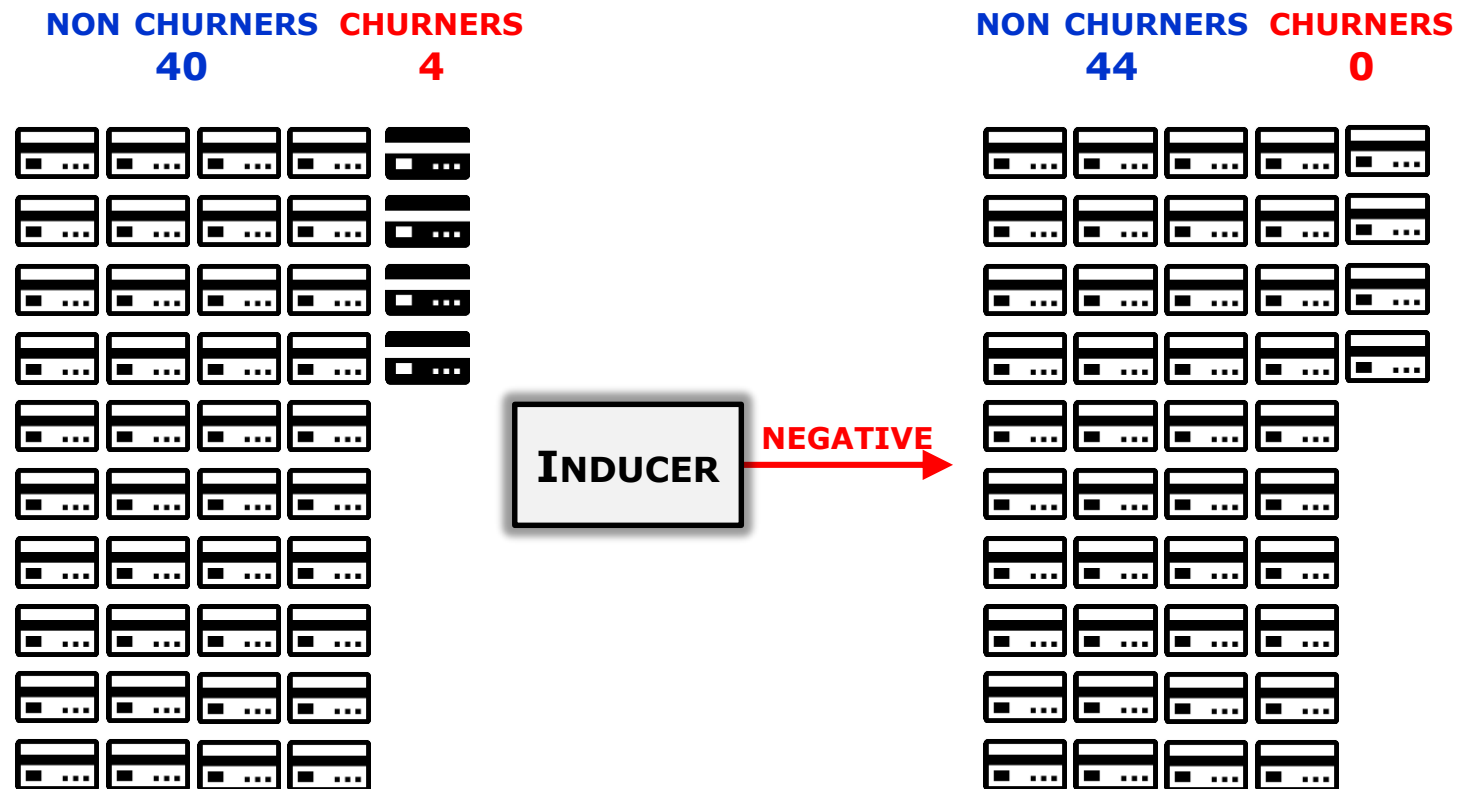
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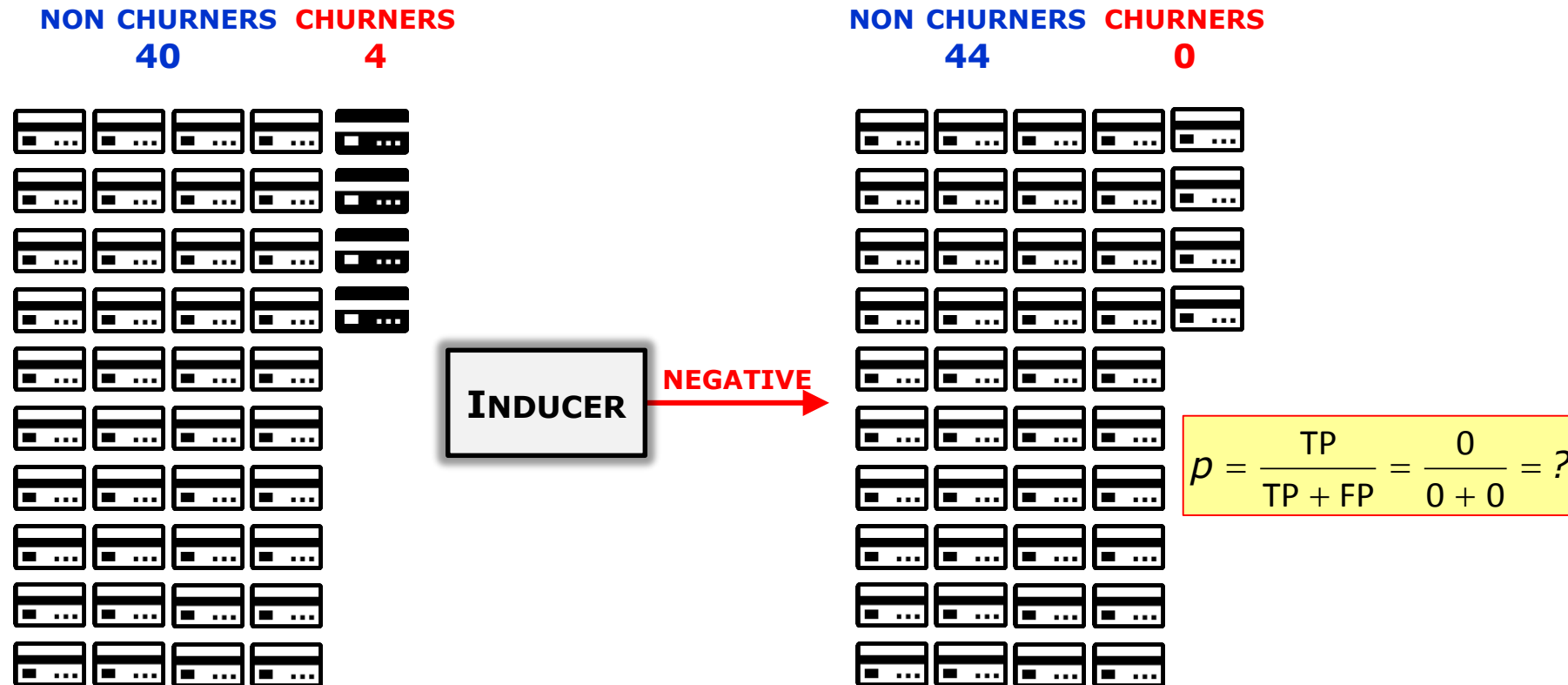
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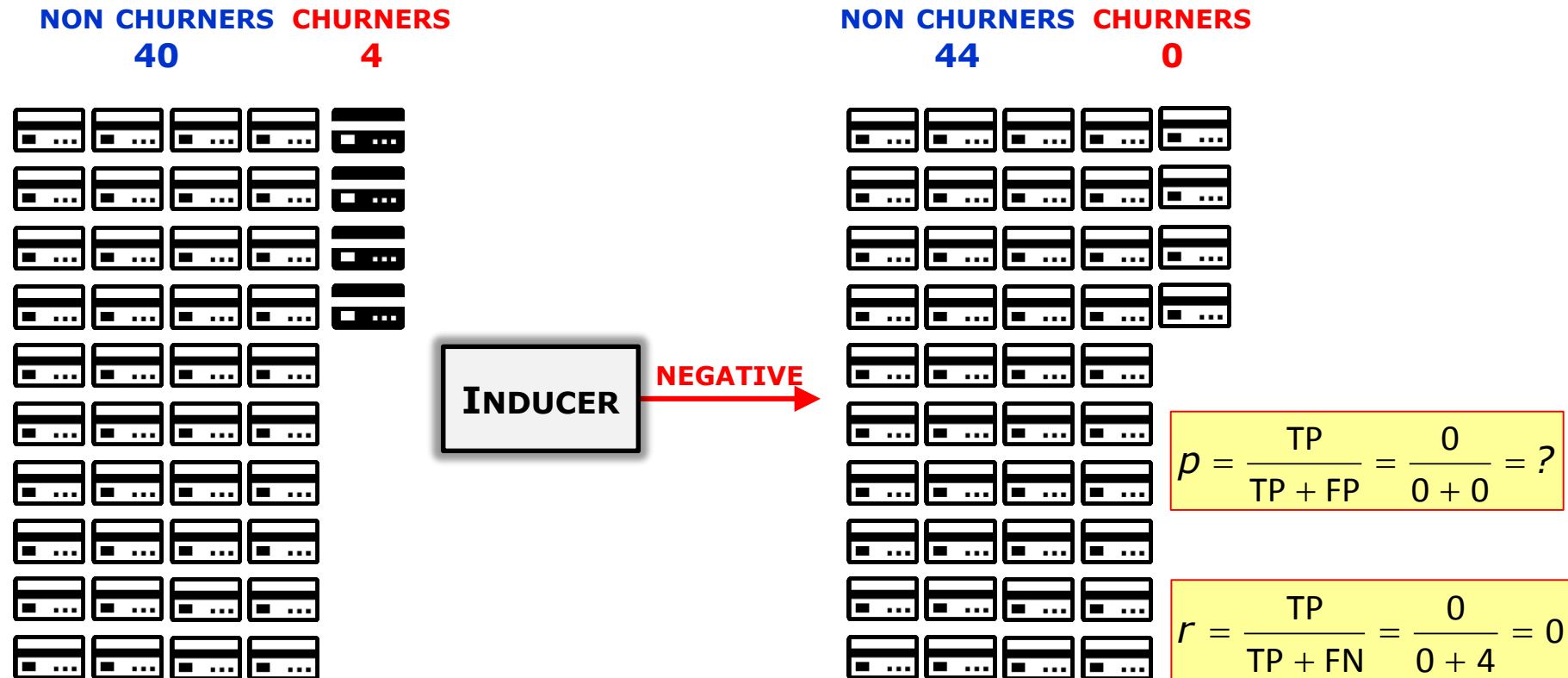
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It is often possible to construct baseline models that maximize one metric but not the other.

PRECISION AND RECALL are usually **SUMMARIZED** into another metric known as the **F_1 MEASURE**:

$$F_1 = \frac{2 \cdot r \cdot p}{r + p}$$

The **F_1 MEASURE** is the **HARMONIC MEAN BETWEEN RECALL AND PRECISION**.

A high value of **F_1 MEASURE** ensures that both **RECALL AND PRECISION** are reasonably high.

A generalization exists under the name **F_β MEASURE** to examine the tradeoff between **RECALL AND PRECISION**:

$$F_\beta = \frac{(\beta^2 + 1) \cdot r \cdot p}{r + \beta^2 \cdot p}$$

F_β with **$\beta=0$** is the **PRECISION**

F_β with **$\beta=\infty$** is the **RECALL**